Status	Finished
Started	Sunday, 11 October 2026, 8:38 PM
Completed	Sunday, 11 October 2026, 8:45 PM
Duration	7 mins 21 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search can find multiple instances of the target element
b. Binary search does not require dividing the list
c. Binary search works on unsorted lists
d. Binary search has a lower time complexity on large, sorted lists ✓

Question **2**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search continues in the right sublist
b. The list is sorted
c. The search ends successfully ✓
d. The search continues in the left sublist

Question **3**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(1) ✓
b. O👎
c. O(log n)
d. O(n log n)

Question **4**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Selection

b. Complexity

c. Bubble ✓

d. Insertion

Question **5**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By sorting the list first

b. By comparing each element in the list sequentially ✓

c. By dividing the list into halves

d. By using a hash function

Question **6**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Jump search

b. Hashing

c. Binary search

d. Linear search ✓

Question **7**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Discovering

b. Searching ✓

c. Finding

d. Mining

Question **8**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. n ✓

b. log n

c. 1

d. n/2

Question **9**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list contains duplicate elements

b. When the list is sorted ✓

c. When the list is unsorted

d. When the list is very small

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search requires the list to be sorted.

b. Linear search can be applied to both sorted and unsorted lists. ✓

c. Linear search is more efficient than binary search on large lists.

d. Linear search divides the list into halves.

Question **11**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

- a. When the element is found in the list ✓
- b. When the search algorithm finishes
- c. When the list is sorted
- d. When the list contains duplicate elements

Question **12**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

- a. Hash search
- b. Sequential search
- c. Linear search
- d. Binary search ✓

Question **13**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

- a. The lower middle element is chosen as the middle element ✗
- b. The search stops
- c. The middle element is chosen randomly
- d. The higher middle element is chosen as the middle element

Question **14**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

- a. O(1)
- b. O(log n)
- c. O👎 ✓
- d. O(n log n)

Question **15**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

- a. When the list is small and unsorted ✓
- b. When the list is large and sorted
- c. When the list is small and sorted
- d. When the list is large and unsorted